

EQUIVALENT REPRESENTATIONS ARE NOT EQUIVALENT DISTILLATION TARGETS

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ABSTRACT

Embedding distillation across different teacher and student dimensions is commonly handled with learned cross-space projectors, intermediate-layer alignment, or relational objectives. These approaches either optimize through an auxiliary interface that does not uniquely constrain the geometry of the deployed student embedding, require additional alignment machinery and correspondence choices, or discard concrete coordinate supervision altogether. We instead view target construction as two separate decisions: reducing the teacher representation to the student dimension and choosing how that representation should be expressed to the pretrained student. GATE-KD makes this separation explicit. It constructs a student-width teacher representation and selects a student-compatible coordinate realization in closed form, directly supervising the deployed embedding without a trainable cross-space interface or intermediate-layer correspondence. The distinction is substantial: with retained teacher information and relational geometry held fixed, orthogonally equivalent targets differ by 4.93 points across nine benchmarks. Across three teacher–student configurations, two distillation corpora, and zero-shot retrieval, GATE-KD achieves the highest average in eight of nine evaluated settings, improves over the strongest baseline by up to 1.59 points, and reduces measured optimization-loop time by up to 4.0×. GATE-KD thus achieves stronger embedding distillation with less alignment machinery.

1 METHODOLOGY

Let $f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}$ be a frozen teacher and $f_S(\cdot; \theta) : \mathcal{X} \rightarrow \mathbb{R}^{d_S}$ a pretrained student. Given an unlabeled distillation corpus $\mathcal{C} = \{x_i\}_{i=1}^N$, we denote their normalized embeddings by

$$\mathbf{t}_i = \text{norm}(f_T(x_i)), \quad \mathbf{z}_i(\theta) = \text{norm}(f_S(x_i; \theta)), \quad (1)$$

where $\mathbf{z}_i$ is the student embedding used directly at inference.

When $d_T \neq d_S$, embedding distillation must resolve two separate choices. The first is *dimension reduction*: which teacher representation should be retained at width d_S . The second is *coordinate selection*: once that representation has been reduced, which coordinate realization should be presented to the pretrained student. GATE-KD treats these choices explicitly rather than learning a cross-space interface that couples them.

1.1 DIMENSION REDUCTION DOES NOT DETERMINE A DISTILLATION TARGET

We first construct a fixed teacher representation at the student width. In our implementation, when $d_T > d_S$, we fit the leading d_S PCA directions $P \in \mathbb{R}^{d_T \times d_S}$ on the centered teacher cache and define

$$\mathbf{y}_i = \text{norm}(\mathbf{t}_i P). \quad (2)$$

Centering is used only to estimate the PCA directions: the mean is not subtracted from $\mathbf{t}_i P$, and the retained components are not whitened. PCA is only our default dimension-reduction operator. GATE-KD applies to any fixed procedure that produces a student-width teacher matrix $Y \in \mathbb{R}^{N \times d_S}$.

Crucially, constructing Y does not finish target construction. For every orthogonal matrix $R \in O(d_S)$,

$$(YR)(YR)^\top = YY^\top. \quad (3)$$

Thus,

$$[Y]_{O(d_S)} = \{YR : R \in O(d_S)\} \quad (4)$$

contains an entire family of reduced teacher representations with identical pairwise inner products and Euclidean distances. Dimension reduction determines the relational geometry to retain, but not the concrete coordinates in which that geometry is presented to the student.

This remaining choice matters whenever distillation directly supervises the student endpoint. For a fixed student matrix $Z \in \mathbb{R}^{N \times d_S}$, consider the mean cosine loss toward YR ,

$$\begin{aligned} \mathcal{L}_Z(R) &= \frac{1}{N} \sum_{i=1}^N (1 - \langle \mathbf{z}_i, \mathbf{y}_i R \rangle) \\ &= 1 - \frac{1}{N} \text{tr}(R^\top Y^\top Z). \end{aligned} \quad (5)$$

Although every YR has the same relational geometry, its loss against the pretrained student depends on R . In particular,

$$\min_{R \in O(d_S)} \mathcal{L}_Z(R) = 1 - \frac{\|Y^\top Z\|_*}{N}, \quad (6)$$

where $\|\cdot\|_*$ denotes the nuclear norm. Hence geometric equivalence after dimension reduction does not imply equivalence as an optimization target. The remaining problem is to choose a coordinate realization of Y relative to the student that will actually be optimized.

1.2 STUDENT-CONDITIONED COORDINATE SELECTION

GATE-KD makes coordinate selection an explicit second stage. Given the reduced teacher representation Y and the current student state Z , we choose

$$R^* = \arg \min_{R \in O(d_S)} \|YR - Z\|_F^2. \quad (7)$$

Because the rows of Y and Z are normalized, this objective is equivalent to minimizing the aggregate cosine endpoint loss in Equation 5. Given

$$Y^\top Z = U \Sigma V^\top, \quad (8)$$

the solution is available in closed form (Schönemann, 1966):

$$R^* = UV^\top, \quad \boldsymbol{\tau}_i = \mathbf{y}_i R^*. \quad (9)$$

The transformation changes only the coordinates of the reduced representation: YR^* preserves exactly the same Gram matrix as Y . Its role is therefore not to retain additional teacher information, but to choose how the retained information is presented to this particular pretrained student. The selected coordinates are student-dependent rather than canonical teacher coordinates.

We estimate this transformation at the *corpus level* rather than independently for each minibatch. Let $\mathcal{G} \subseteq \mathcal{C}$ denote a representative fit set. At refresh step r , we obtain normalized student embeddings $Z_{\mathcal{G}}^{(r)}$ in evaluation mode and without gradients, and compute

$$Y_{\mathcal{G}}^\top Z_{\mathcal{G}}^{(r)} = U \Sigma V^\top, \quad R^{(r)} = UV^\top. \quad (10)$$

The same $R^{(r)}$ is then used for every minibatch until the next refresh,

$$\boldsymbol{\tau}_i^{(r)} = \mathbf{y}_i R^{(r)}. \quad (11)$$

For a minibatch $\mathcal{B}$, the endpoint objective is

$$\mathcal{L}_{\text{end}}^{(r)} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left(1 - \langle \mathbf{z}_i, \boldsymbol{\tau}_i^{(r)} \rangle \right). \quad (12)$$

A single corpus-level transformation is important because independently choosing R for different minibatches preserves geometry only within each minibatch; their relative transformations need not

define any single representation of the full teacher corpus. A shared $R^{(r)}$, in contrast, preserves the complete teacher Gram matrix while providing one persistent endpoint target.

The representative fitting step is outside backpropagation. No gradients pass through the student embeddings used to estimate $R^{(r)}$, the SVD, or the cached teacher targets. Our default recipe selects $R^{(0)}$ from the pretrained student and refreshes it after each non-final epoch. Ordinary student updates therefore optimize directly against a fixed target between refreshes, without introducing a trainable teacher-to-student projector or modifying the embedding used at inference.

1.3 COORDINATE-INVARIANT STRUCTURAL SUPPORT

Coordinate selection determines the sample-specific endpoint target, but it does not add relational information beyond that already contained in the reduced teacher representation. We therefore complement the endpoint objective with a structural signal computed directly in the native teacher and student spaces. The auxiliary term is required to be invariant to orthogonal transformations, so that it constrains geometry without introducing a second, competing coordinate preference.

For normalized embeddings, we use chord distance

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u}, \mathbf{v} \rangle}. \quad (13)$$

Within each minibatch of size B , we independently construct minimum spanning trees over the native teacher embeddings $\{\mathbf{t}_i\}$ and current student embeddings $\{\mathbf{z}_i\}$. The $B - 1$ finite H_0 persistence death scales coincide with the MST edge weights (Hofer et al., 2020). Let

$$q_{(1)}^T \leq \dots \leq q_{(B-1)}^T, \quad q_{(1)}^S \leq \dots \leq q_{(B-1)}^S \quad (14)$$

be the sorted teacher and student edge weights. We define

$$\mathcal{L}_{H_0} = \frac{1}{B-1} \sum_{j=1}^{B-1} \left(q_{(j)}^S - q_{(j)}^T \right)^2. \quad (15)$$

The two trees are constructed independently in their native spaces, so this term requires neither equal embedding widths nor shared coordinates. Because orthogonal transformations preserve inner products, they also preserve chord distances, MSTs, and their edge weights. Consequently, $\mathcal{L}_{H_0}$ is invariant to the same coordinate choice resolved by the endpoint term.

The two objectives therefore have separate roles. $\mathcal{L}_{\text{end}}$ selects and supervises a concrete coordinate realization relative to the student, whereas $\mathcal{L}_{H_0}$ constrains native-space relational structure without preferring any realization. The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{end}} + \lambda_{H_0} \mathcal{L}_{H_0} \quad (16)$$

Teacher embeddings and the dimension-reduction operator are computed once. Each coordinate refresh requires only a no-gradient student pass over the fit set and one $d_S \times d_S$ SVD; ordinary minibatch updates require no online teacher forward pass. All target construction and structural computation are removed after training, leaving only the student encoder and its normalized endpoint at inference.

Table 1: Teacher–student interface choices in configuration (c). Learned-map rows are carried over from the matched interface sweep; fixed-target rows use the current ablation suite. Results are mean $\pm$ sample standard deviation over three seeds; the best result is bold.

Interface strategy	Trainable map?	Overall $\uparrow$
Learned teacher-to-student	Yes	61.46 $\pm$ 0.39
Learned student-to-teacher	Yes	69.33 $\pm$ 0.32
PCA-reduced target	No	72.39 $\pm$ 0.03
Haar-random target	No	73.01 $\pm$ 0.27
Student-conditioned target	No	74.86 $\pm$ 0.06

Table 2: Dimension reduction and coordinate selection in configuration (c). Results are mean $\pm$ sample standard deviation over three seeds. Random rows are independent subspace draws. The best coordinate treatment within each reduction is bold.

Reduction	Energy $\uparrow$	Overall $\uparrow$		
		As reduced	Haar-random	Student-conditioned
PCA-64	0.593	71.49 $\pm$ 0.08	72.39 $\pm$ 0.13	73.70 $\pm$ 0.10
PCA-128	0.739	72.22 $\pm$ 0.04	73.03 $\pm$ 0.22	74.52 $\pm$ 0.01
PCA-256	0.873	72.48 $\pm$ 0.01	73.08 $\pm$ 0.30	74.79 $\pm$ 0.03
PCA-384	0.928	72.39 $\pm$ 0.03	73.01 $\pm$ 0.27	74.86 $\pm$ 0.06
Random subspace (1)	0.381	73.10 $\pm$ 0.08	72.79 $\pm$ 0.32	74.50 $\pm$ 0.03
Random subspace (2)	0.376	72.92 $\pm$ 0.11	72.85 $\pm$ 0.33	74.64 $\pm$ 0.04
Random subspace (3)	0.374	72.65 $\pm$ 0.11	72.85 $\pm$ 0.16	74.47 $\pm$ 0.07
SVD (uncentered)	0.901	72.54 $\pm$ 0.07	73.11 $\pm$ 0.15	74.46 $\pm$ 0.03

Table 3: Controls for student-conditioned coordinate selection in configuration (c). Panel (a) varies the signal used to select the coordinates; panel (b) varies how often the representative is fitted. All rows use the same rank-384 reduction (0.928 retained energy). Results are mean $\pm$ sample standard deviation over three seeds; the best value in each panel is bold.

Treatment	Source $\uparrow$	Non-source $\uparrow$	Overall $\uparrow$
<i>(a) Selection signal</i>			
As reduced	65.64 $\pm$ 0.04	75.77 $\pm$ 0.06	72.39 $\pm$ 0.03
Haar-random	66.82 $\pm$ 0.44	76.11 $\pm$ 0.18	73.01 $\pm$ 0.27
Shuffled correspondence	67.57 $\pm$ 0.36	77.23 $\pm$ 0.20	74.01 $\pm$ 0.21
Unrelated student	67.28 $\pm$ 0.47	76.08 $\pm$ 0.06	73.15 $\pm$ 0.20
Matched student	68.80 $\pm$ 0.16	77.89 $\pm$ 0.01	74.86 $\pm$ 0.06
<i>(b) Representative schedule</i>			
As reduced	65.64 $\pm$ 0.04	75.77 $\pm$ 0.06	72.39 $\pm$ 0.03
Haar-random	66.82 $\pm$ 0.44	76.11 $\pm$ 0.18	73.01 $\pm$ 0.27
Per-batch selection	65.75 $\pm$ 0.20	76.19 $\pm$ 0.09	72.71 $\pm$ 0.07
Initial selection only	68.24 $\pm$ 0.10	77.80 $\pm$ 0.04	74.61 $\pm$ 0.04
One refresh	68.34 $\pm$ 0.02	77.85 $\pm$ 0.03	74.68 $\pm$ 0.01
Epoch-wise selection	68.80 $\pm$ 0.16	77.89 $\pm$ 0.01	74.86 $\pm$ 0.06

Table 4: Nine-task results on the 15K distillation corpus. Distilled students report mean $\pm$ sample standard deviation over three seeds; best and second-best results are bold and underlined. GATE-KD uses $\lambda_{H_0} = 0.5$.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Averages		
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	Src	Non-src	All
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)												
Teacher	93.26	71.04	69.08	84.80	91.77	66.79	83.43	82.55	86.87	74.25	84.48	81.07
Student (pre-KD)	84.33	67.08	47.79	77.36	67.74	59.22	42.43	21.54	20.29	42.43	60.08	54.20
RKD	89.76 $\pm$ 0.11	70.90 $\pm$ 0.31	59.98 $\pm$ 0.05	81.94 $\pm$ 0.26	75.59 $\pm$ 0.13	69.02 $\pm$ 0.18	72.93 $\pm$ 0.07	67.84 $\pm$ 0.18	73.72 $\pm$ 0.08	67.57 $\pm$ 0.05	76.49 $\pm$ 0.07	73.52 $\pm$ 0.06
Stella	88.48 $\pm$ 0.30	68.94 $\pm$ 0.17	53.69 $\pm$ 0.50	83.95 $\pm$ 0.10	75.39 $\pm$ 0.54	71.93 $\pm$ 0.09	69.87 $\pm$ 0.64	64.39 $\pm$ 0.21	70.86 $\pm$ 0.36	65.49 $\pm$ 0.17	75.17 $\pm$ 0.13	71.95 $\pm$ 0.06
CDM	88.13 $\pm$ 0.23	69.03 $\pm$ 0.20	52.83 $\pm$ 0.50	83.00 $\pm$ 0.06	77.10 $\pm$ 0.27	71.84 $\pm$ 0.09	67.89 $\pm$ 0.37	61.06 $\pm$ 0.37	69.19 $\pm$ 0.42	64.62 $\pm$ 0.28	74.37 $\pm$ 0.12	71.12 $\pm$ 0.18
DSKD	87.97 $\pm$ 0.16	68.90 $\pm$ 0.28	53.07 $\pm$ 0.31	83.03 $\pm$ 0.15	76.98 $\pm$ 0.59	<u>71.86</u> $\pm$ 0.08	67.77 $\pm$ 0.66	61.11 $\pm$ 0.26	69.55 $\pm$ 0.12	64.83 $\pm$ 0.12	74.29 $\pm$ 0.14	71.14 $\pm$ 0.10
EMO	86.89 $\pm$ 0.07	68.99 $\pm$ 0.11	52.02 $\pm$ 1.09	79.74 $\pm$ 0.12	76.37 $\pm$ 0.35	69.62 $\pm$ 0.16	67.89 $\pm$ 0.09	56.46 $\pm$ 0.47	64.54 $\pm$ 0.50	62.06 $\pm$ 0.49	72.72 $\pm$ 0.15	69.17 $\pm$ 0.24
TALAS	<u>90.53</u> $\pm$ 0.09	<u>73.77</u> $\pm$ 0.21	<u>68.47</u> $\pm$ 0.34	85.38 $\pm$ 0.05	81.83 $\pm$ 0.21	69.59 $\pm$ 0.14	<u>76.79</u> $\pm$ 0.06	<u>70.79</u> $\pm$ 0.19	78.50 $\pm$ 0.08	72.19 $\pm$ 0.07	<u>79.85</u> $\pm$ 0.07	<u>77.29</u> $\pm$ 0.04
GATE-KD	90.60 $\pm$ 0.21	73.86 $\pm$ 0.10	69.48 $\pm$ 0.22	<u>85.31</u> $\pm$ 0.11	<u>80.23</u> $\pm$ 0.13	68.19 $\pm$ 0.12	77.80 $\pm$ 0.14	75.15 $\pm$ 0.07	<u>78.44</u> $\pm$ 0.16	<u>72.04</u> $\pm$ 0.08	80.49 $\pm$ 0.03	77.67 $\pm$ 0.03
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)												
Teacher	93.49	73.75	69.03	85.80	91.87	61.46	79.18	78.74	84.86	71.78	83.81	79.80
Student (pre-KD)	81.18	71.30	48.85	79.66	65.06	60.89	49.46	26.76	24.12	44.62	62.24	56.36
RKD	89.14 $\pm$ 0.09	73.29 $\pm$ 0.09	<u>62.23</u> $\pm$ 0.27	84.35 $\pm$ 0.06	78.74 $\pm$ 0.11	65.02 $\pm$ 0.10	73.20 $\pm$ 0.19	65.02 $\pm$ 0.29	71.97 $\pm$ 0.38	66.41 $\pm$ 0.07	77.29 $\pm$ 0.06	73.66 $\pm$ 0.04
Stella	86.82 $\pm$ 0.28	70.06 $\pm$ 0.12	56.61 $\pm$ 0.74	83.02 $\pm$ 0.12	74.63 $\pm$ 0.12	69.72 $\pm$ 0.05	69.08 $\pm$ 0.20	63.14 $\pm$ 0.28	67.10 $\pm$ 0.10	64.47 $\pm$ 0.27	74.46 $\pm$ 0.05	71.13 $\pm$ 0.11
CDM	87.82 $\pm$ 0.19	69.99 $\pm$ 0.09	58.32 $\pm$ 0.51	83.44 $\pm$ 0.07	75.56 $\pm$ 0.40	<u>69.70</u> $\pm$ 0.02	65.93 $\pm$ 0.16	60.84 $\pm$ 0.34	65.31 $\pm$ 0.15	64.45 $\pm$ 0.13	73.93 $\pm$ 0.06	70.77 $\pm$ 0.02
DSKD	87.70 $\pm$ 0.16	70.35 $\pm$ 0.32	59.71 $\pm$ 0.61	82.99 $\pm$ 0.18	75.58 $\pm$ 0.12	69.59 $\pm$ 0.04	65.36 $\pm$ 0.37	61.21 $\pm$ 0.31	65.02 $\pm$ 0.18	64.77 $\pm$ 0.27	73.87 $\pm$ 0.10	70.83 $\pm$ 0.11
EMO	87.54 $\pm$ 0.18	71.03 $\pm$ 0.14	58.52 $\pm$ 0.30	82.46 $\pm$ 0.18	74.73 $\pm$ 0.58	69.20 $\pm$ 0.22	64.71 $\pm$ 0.55	56.89 $\pm$ 0.27	61.25 $\pm$ 0.03	62.99 $\pm$ 0.19	72.89 $\pm$ 0.10	69.59 $\pm$ 0.12
TALAS	<u>89.83</u> $\pm$ 0.09	74.70 $\pm$ 0.09	61.65 $\pm$ 0.23	86.62 $\pm$ 0.05	<u>80.85</u> $\pm$ 0.22	66.72 $\pm$ 0.07	<u>75.40</u> $\pm$ 0.10	<u>67.08</u> $\pm$ 0.31	<u>77.08</u> $\pm$ 0.01	<u>68.48</u> $\pm$ 0.07	<u>79.08</u> $\pm$ 0.08	<u>75.55</u> $\pm$ 0.06
GATE-KD	90.17 $\pm$ 0.05	<u>73.74</u> $\pm$ 0.24	64.83 $\pm$ 0.72	<u>86.42</u> $\pm$ 0.09	80.97 $\pm$ 0.08	65.47 $\pm$ 0.09	76.77 $\pm$ 0.03	68.27 $\pm$ 0.04	77.63 $\pm$ 0.12	69.31 $\pm$ 0.21	79.39 $\pm$ 0.06	76.03 $\pm$ 0.11
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)												
Teacher	92.51	71.28	65.80	83.38	90.09	66.84	80.65	77.11	84.59	72.41	82.50	79.14
Student (pre-KD)	67.03	69.45	42.90	78.39	64.55	59.58	47.60	21.96	22.08	41.52	58.16	52.62
RKD	<u>86.76</u> $\pm$ 0.07	69.87 $\pm$ 0.21	56.31 $\pm$ 0.35	81.70 $\pm$ 0.03	76.03 $\pm$ 0.20	<u>68.56</u> $\pm$ 0.22	69.21 $\pm$ 0.18	64.26 $\pm$ 0.19	70.25 $\pm$ 0.21	65.04 $\pm$ 0.20	74.64 $\pm$ 0.09	71.44 $\pm$ 0.10
Stella	84.04 $\pm$ 0.04	68.73 $\pm$ 0.44	52.41 $\pm$ 0.28	81.50 $\pm$ 0.12	72.26 $\pm$ 0.32	68.65 $\pm$ 0.16	64.64 $\pm$ 0.19	60.59 $\pm$ 0.12	64.27 $\pm$ 0.20	61.78 $\pm$ 0.13	71.96 $\pm$ 0.13	68.57 $\pm$ 0.10
CDM	84.74 $\pm$ 0.21	69.52 $\pm$ 0.20	53.76 $\pm$ 0.95	81.15 $\pm$ 0.14	72.86 $\pm$ 0.30	68.29 $\pm$ 0.11	63.98 $\pm$ 0.36	60.43 $\pm$ 0.22	63.52 $\pm$ 0.24	61.86 $\pm$ 0.43	72.11 $\pm$ 0.19	68.69 $\pm$ 0.27
DSKD	84.76 $\pm$ 0.10	69.56 $\pm$ 0.20	53.14 $\pm$ 0.29	81.15 $\pm$ 0.01	72.81 $\pm$ 0.09	68.33 $\pm$ 0.04	63.78 $\pm$ 0.06	60.24 $\pm$ 0.19	63.53 $\pm$ 0.18	61.67 $\pm$ 0.03	72.05 $\pm$ 0.06	68.59 $\pm$ 0.04
EMO	83.54 $\pm$ 0.31	69.31 $\pm$ 0.06	54.48 $\pm$ 0.27	80.87 $\pm$ 0.13	72.00 $\pm$ 0.16	67.89 $\pm$ 0.09	62.19 $\pm$ 0.16	57.04 $\pm$ 0.17	61.24 $\pm$ 0.23	61.20 $\pm$ 0.09	70.82 $\pm$ 0.07	67.62 $\pm$ 0.02
TALAS	86.39 $\pm$ 0.13	<u>71.81</u> $\pm$ 0.11	<u>58.40</u> $\pm$ 0.30	84.25 $\pm$ 0.03	<u>79.29</u> $\pm$ 0.11	66.82 $\pm$ 0.02	<u>70.42</u> $\pm$ 0.10	<u>68.08</u> $\pm$ 0.22	<u>73.34</u> $\pm$ 0.10	<u>66.19</u> $\pm$ 0.11	<u>76.71</u> $\pm$ 0.04	<u>73.20</u> $\pm$ 0.06
GATE-KD	87.77 $\pm$ 0.07	72.30 $\pm$ 0.19	63.45 $\pm$ 0.43	<u>84.19</u> $\pm$ 0.03	81.56 $\pm$ 0.16	68.17 $\pm$ 0.05	72.59 $\pm$ 0.03	68.91 $\pm$ 0.03	74.79 $\pm$ 0.05	68.80 $\pm$ 0.16	77.89 $\pm$ 0.01	74.86 $\pm$ 0.06

Table 5: Ablation of endpoint targets and structural objectives in configuration (c). Results are mean $\pm$ sample standard deviation over three seeds. Δ is measured against the student-conditioned target without a structural objective; best values are bold.

Endpoint target	Structural objective	Source $\uparrow$	Non-source $\uparrow$	Overall $\uparrow$	Δ
No endpoint	None	41.61 $\pm$ 0.00	58.19 $\pm$ 0.03	52.66 $\pm$ 0.02	—
No endpoint	H_0 persistence	57.55 $\pm$ 0.47	67.53 $\pm$ 0.60	64.21 $\pm$ 0.48	—
PCA-reduced	None	62.35 $\pm$ 0.22	72.76 $\pm$ 0.04	69.29 $\pm$ 0.07	—
Student-conditioned	None	67.61 $\pm$ 0.09	77.57 $\pm$ 0.06	74.25 $\pm$ 0.01	—
Student-conditioned	Gram matching	67.64 $\pm$ 0.15	76.94 $\pm$ 0.05	73.84 $\pm$ 0.03	-0.41
Student-conditioned	H_0 persistence	68.80 $\pm$ 0.16	77.89 $\pm$ 0.01	74.86 $\pm$ 0.06	+0.61
Student-conditioned	kNN-distance matching	68.79 $\pm$ 0.14	78.11 $\pm$ 0.03	75.01 $\pm$ 0.02	+0.76

Table 6: Zero-shot BEIR retrieval (nDCG@10). Distilled students report mean $\pm$ standard deviation over three seeds; best and second-best results are bold and underlined.

Method	ArguAna	SciFact	SciDocs	NFCorpus	FiQA-2018	Avg.
<i>(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)</i>						
Teacher	74.66	73.75	25.69	36.23	50.15	52.10
Student (pre-KD)	12.65	0.89	0.38	1.33	0.05	3.06
RKD	39.68 $\pm$ 0.25	9.74 $\pm$ 0.64	1.19 $\pm$ 0.04	4.30 $\pm$ 0.30	2.82 $\pm$ 0.09	11.55 $\pm$ 0.18
Stella	39.12 $\pm$ 0.17	14.29 $\pm$ 0.17	<u>3.79</u> $\pm$ 0.26	6.21 $\pm$ 0.14	5.57 $\pm$ 0.04	13.79 $\pm$ 0.03
CDM	33.62 $\pm$ 0.27	15.11 $\pm$ 0.20	3.28 $\pm$ 0.14	6.26 $\pm$ 0.09	5.08 $\pm$ 0.22	12.67 $\pm$ 0.04
DSKD	33.70 $\pm$ 0.39	15.75 $\pm$ 0.49	3.35 $\pm$ 0.11	6.21 $\pm$ 0.17	5.34 $\pm$ 0.24	12.87 $\pm$ 0.27
EMO	21.90 $\pm$ 0.34	8.84 $\pm$ 0.28	1.87 $\pm$ 0.11	4.87 $\pm$ 0.16	2.04 $\pm$ 0.06	7.91 $\pm$ 0.07
TALAS	48.95 $\pm$ 0.17	19.94 $\pm$ 0.09	5.70 $\pm$ 0.12	7.94 $\pm$ 0.08	8.15 $\pm$ 0.02	18.14 $\pm$ 0.07
GATE-KD	<u>45.20</u> $\pm$ 0.32	<u>19.25</u> $\pm$ 0.66	2.81 $\pm$ 0.09	<u>6.79</u> $\pm$ 0.22	<u>6.56</u> $\pm$ 0.21	<u>16.12</u> $\pm$ 0.26
<i>(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)</i>						
Teacher	54.01	64.16	16.37	31.80	41.12	41.49
Student (pre-KD)	9.32	1.15	0.44	1.22	0.08	2.44
RKD	21.27 $\pm$ 0.23	<u>7.84</u> $\pm$ 0.12	<u>1.96</u> $\pm$ 0.05	<u>4.31</u> $\pm$ 0.14	2.26 $\pm$ 0.03	7.53 $\pm$ 0.09
Stella	18.46 $\pm$ 0.37	3.11 $\pm$ 0.27	1.25 $\pm$ 0.06	2.29 $\pm$ 0.15	1.66 $\pm$ 0.08	5.35 $\pm$ 0.16
CDM	15.83 $\pm$ 0.19	3.77 $\pm$ 0.38	0.88 $\pm$ 0.02	2.88 $\pm$ 0.16	1.41 $\pm$ 0.21	4.95 $\pm$ 0.15
DSKD	16.25 $\pm$ 0.15	4.05 $\pm$ 0.10	0.88 $\pm$ 0.03	2.96 $\pm$ 0.11	1.59 $\pm$ 0.07	5.15 $\pm$ 0.07
EMO	13.10 $\pm$ 0.27	2.83 $\pm$ 0.17	0.72 $\pm$ 0.04	2.25 $\pm$ 0.07	0.85 $\pm$ 0.09	3.95 $\pm$ 0.08
TALAS	30.14 $\pm$ 0.32	6.55 $\pm$ 0.34	1.90 $\pm$ 0.08	2.82 $\pm$ 0.18	<u>2.83</u> $\pm$ 0.09	<u>8.85</u> $\pm$ 0.17
GATE-KD	<u>25.30</u> $\pm$ 0.02	11.22 $\pm$ 0.27	3.31 $\pm$ 0.02	4.93 $\pm$ 0.06	4.83 $\pm$ 0.01	9.92 $\pm$ 0.05
<i>(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)</i>						
Teacher	68.11	68.26	15.72	29.73	38.68	44.10
Student (pre-KD)	9.85	0.66	0.32	1.28	0.05	2.43
RKD	30.12 $\pm$ 0.79	<u>5.57</u> $\pm$ 0.76	<u>1.77</u> $\pm$ 0.12	<u>3.34</u> $\pm$ 0.07	<u>1.88</u> $\pm$ 0.06	8.54 $\pm$ 0.33
Stella	24.49 $\pm$ 0.45	2.65 $\pm$ 0.09	1.09 $\pm$ 0.07	2.62 $\pm$ 0.07	1.01 $\pm$ 0.11	6.37 $\pm$ 0.09
CDM	21.09 $\pm$ 0.30	1.66 $\pm$ 0.17	1.13 $\pm$ 0.03	2.57 $\pm$ 0.05	0.89 $\pm$ 0.02	5.47 $\pm$ 0.06
DSKD	21.56 $\pm$ 0.38	1.77 $\pm$ 0.20	1.18 $\pm$ 0.06	2.63 $\pm$ 0.03	0.89 $\pm$ 0.07	5.61 $\pm$ 0.13
EMO	17.18 $\pm$ 0.14	1.41 $\pm$ 0.09	0.92 $\pm$ 0.02	1.96 $\pm$ 0.06	0.70 $\pm$ 0.03	4.44 $\pm$ 0.05
TALAS	<u>35.21</u> $\pm$ 0.52	4.17 $\pm$ 0.09	1.44 $\pm$ 0.04	3.12 $\pm$ 0.10	1.87 $\pm$ 0.07	<u>9.16</u> $\pm$ 0.14
GATE-KD	36.18 $\pm$ 0.09	10.26 $\pm$ 0.21	3.44 $\pm$ 0.06	4.30 $\pm$ 0.06	3.57 $\pm$ 0.03	11.55 $\pm$ 0.06

Table 7: Optimization-loop cost on NVIDIA H200 GPUs, excluding non-training overhead. Results are mean $\pm$ standard deviation over three seeds; best and second-best values are bold and underlined.

Method	Overall $\uparrow$	ms/step $\downarrow$	samples/s $\uparrow$	Minutes $\downarrow$	Peak GiB $\downarrow$
<i>(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)</i>					
RKD	73.52 $\pm$ 0.06	119.7 $\pm$ 18.3	1081.0 $\pm$ 179.8	1.20 $\pm$ 0.18	1.72 $\pm$ 0.00
Stella	71.95 $\pm$ 0.06	130.7 $\pm$ 8.1	975.9 $\pm$ 58.8	1.27 $\pm$ 0.08	1.78 $\pm$ 0.01
CDM	71.12 $\pm$ 0.18	340.4 $\pm$ 13.6	374.2 $\pm$ 15.3	3.38 $\pm$ 0.12	9.31 $\pm$ 0.01
DSKD	71.14 $\pm$ 0.10	290.9 $\pm$ 9.0	437.7 $\pm$ 13.3	2.88 $\pm$ 0.09	9.45 $\pm$ 0.00
EMO	69.17 $\pm$ 0.24	721.1 $\pm$ 12.9	176.5 $\pm$ 3.1	7.02 $\pm$ 0.12	9.30 $\pm$ 0.00
TALAS	77.29 $\pm$ 0.04	329.0 $\pm$ 12.5	387.1 $\pm$ 15.0	3.23 $\pm$ 0.11	1.43 $\pm$ 0.00
GATE-KD	77.67 $\pm$ 0.03	83.4 $\pm$ 3.8	1527.2 $\pm$ 70.8	0.84 $\pm$ 0.04	<u>1.72</u> $\pm$ 0.00
<i>(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)</i>					
RKD	73.66 $\pm$ 0.04	77.5 $\pm$ 8.8	1656.4 $\pm$ 201.2	0.79 $\pm$ 0.09	1.07 $\pm$ 0.00
Stella	71.13 $\pm$ 0.11	152.8 $\pm$ 30.8	859.1 $\pm$ 195.6	1.49 $\pm$ 0.30	1.09 $\pm$ 0.00
CDM	70.77 $\pm$ 0.02	127.5 $\pm$ 13.8	1006.4 $\pm$ 115.8	1.30 $\pm$ 0.12	2.15 $\pm$ 0.01
DSKD	70.83 $\pm$ 0.11	115.0 $\pm$ 21.9	1137.2 $\pm$ 241.0	1.14 $\pm$ 0.20	2.22 $\pm$ 0.01
EMO	69.59 $\pm$ 0.12	394.6 $\pm$ 33.4	324.0 $\pm$ 28.3	3.87 $\pm$ 0.35	2.15 $\pm$ 0.01
TALAS	75.55 $\pm$ 0.06	184.4 $\pm$ 4.6	690.2 $\pm$ 17.3	1.82 $\pm$ 0.03	0.85 $\pm$ 0.00
GATE-KD	76.03 $\pm$ 0.11	56.5 $\pm$ 7.7	2281.6 $\pm$ 310.0	0.57 $\pm$ 0.07	<u>1.07</u> $\pm$ 0.00
<i>(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)</i>					
RKD	71.44 $\pm$ 0.10	64.8 $\pm$ 13.7	2012.3 $\pm$ 488.6	0.68 $\pm$ 0.14	0.41 $\pm$ 0.00
Stella	68.57 $\pm$ 0.10	150.7 $\pm$ 15.3	850.0 $\pm$ 81.5	1.47 $\pm$ 0.13	0.42 $\pm$ 0.00
CDM	68.69 $\pm$ 0.27	161.8 $\pm$ 21.6	796.3 $\pm$ 113.1	1.65 $\pm$ 0.20	1.53 $\pm$ 0.00
DSKD	68.59 $\pm$ 0.04	175.4 $\pm$ 8.2	726.4 $\pm$ 33.5	1.78 $\pm$ 0.06	1.56 $\pm$ 0.00
EMO	67.62 $\pm$ 0.02	499.3 $\pm$ 1.4	254.8 $\pm$ 0.7	4.90 $\pm$ 0.01	1.53 $\pm$ 0.00
TALAS	73.20 $\pm$ 0.06	169.8 $\pm$ 15.4	753.7 $\pm$ 71.4	1.69 $\pm$ 0.14	0.34 $\pm$ 0.00
GATE-KD	74.79 $\pm$ 0.06	62.9 $\pm$ 3.1	2025.2 $\pm$ 98.3	0.64 $\pm$ 0.04	<u>0.41</u> $\pm$ 0.00

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Table 8: Sensitivity analyses for configuration (c). Panel (a) varies the structural weight; panel (b) varies the representative fit-set size. Results are mean $\pm$ sample standard deviation over three seeds; best values in each panel are bold.

Setting	Source $\uparrow$	Non-source $\uparrow$	Overall $\uparrow$
<i>(a) Structural weight λ_{H_0}</i>			
0	67.61 $\pm$ 0.09	77.57 $\pm$ 0.06	74.25 $\pm$ 0.01
0.1	68.13 $\pm$ 0.17	77.75 $\pm$ 0.04	74.54 $\pm$ 0.03
0.3	68.72 $\pm$ 0.28	77.85 $\pm$ 0.06	74.81 $\pm$ 0.05
0.5	68.80 $\pm$ 0.16	77.89 $\pm$ 0.01	74.86 $\pm$ 0.06
1.0	68.38 $\pm$ 0.17	77.75 $\pm$ 0.00	74.63 $\pm$ 0.06
<i>(b) Representative fit-set size</i>			
2,048	68.69 $\pm$ 0.15	77.62 $\pm$ 0.10	74.65 $\pm$ 0.05
4,096	68.79 $\pm$ 0.07	77.77 $\pm$ 0.05	74.78 $\pm$ 0.05
8,192	68.55 $\pm$ 0.09	77.84 $\pm$ 0.04	74.75 $\pm$ 0.01
14,760	68.80 $\pm$ 0.16	77.89 $\pm$ 0.01	74.86 $\pm$ 0.06

Table 9: Nine-task results with the 100K distillation corpus. Distilled students report mean $\pm$ standard deviation over three seeds; best and second-best results are bold and underlined.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Avg.
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)										
Teacher	93.26	71.04	69.08	84.80	91.77	66.79	83.43	82.55	86.87	81.07
Student (pre-KD)	84.33	67.08	47.79	77.36	67.74	59.22	42.43	21.54	20.29	54.20
RKD	92.23	71.96	63.79	85.00	80.27	66.31	79.32	71.72	77.70	76.48
	± 0.15	± 0.18	± 0.40	± 0.04	± 0.38	± 0.10	± 0.02	± 0.16	± 0.20	± 0.04
Stella	90.02	71.90	60.14	85.96	78.82	69.93	76.53	69.79	77.22	75.59
	± 0.15	± 0.22	± 0.45	± 0.05	± 0.27	± 0.13	± 0.23	± 0.12	± 0.16	± 0.10
CDM	89.16	69.63	53.74	85.57	71.22	70.04	71.26	66.97	73.69	72.36
	± 0.04	± 0.23	± 0.13	± 0.08	± 0.19	± 0.05	± 0.29	± 0.21	± 0.38	± 0.13
DSKD	88.87	69.05	54.02	85.90	70.02	69.63	71.07	66.52	73.52	72.07
	± 0.07	± 0.09	± 0.56	± 0.05	± 0.31	± 0.21	± 0.10	± 0.17	± 0.01	± 0.03
EMO	84.96	68.93	54.80	85.05	65.75	68.80	62.00	61.75	69.54	69.07
	± 0.84	± 0.65	± 1.52	± 0.04	± 0.22	± 0.13	± 2.71	± 1.85	± 0.37	± 0.89
TALAS	93.12	74.67	71.42	86.30	87.06	68.09	80.29	76.34	82.26	79.95
	± 0.13	± 0.38	± 0.26	± 0.01	± 0.12	± 0.02	± 0.03	± 0.01	± 0.05	± 0.08
GATE-KD	93.47	74.18	72.50	85.45	87.24	66.57	81.85	76.40	82.52	80.02
	± 0.12	± 0.25	± 0.47	± 0.08	± 0.12	± 0.08	± 0.06	± 0.07	± 0.08	± 0.03
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)										
Teacher	93.49	73.75	69.03	85.80	91.87	61.46	79.18	78.74	84.86	79.80
Student (pre-KD)	81.18	71.30	48.85	79.66	65.06	60.89	49.46	26.76	24.12	56.36
RKD	91.68	74.57	62.45	85.81	79.81	63.57	76.91	69.54	75.56	75.54
	± 0.06	± 0.40	± 0.75	± 0.08	± 0.17	± 0.13	± 0.05	± 0.11	± 0.09	± 0.10
Stella	88.72	73.43	61.63	86.39	74.13	66.98	73.32	68.30	72.66	73.95
	± 0.41	± 0.23	± 0.29	± 0.08	± 0.31	± 0.27	± 0.07	± 0.12	± 0.14	± 0.04
CDM	87.64	70.82	56.37	85.67	67.69	66.89	70.15	66.68	69.49	71.27
	± 0.23	± 0.14	± 0.57	± 0.08	± 0.23	± 0.20	± 0.48	± 0.26	± 0.15	± 0.10
DSKD	86.69	69.76	56.98	84.92	66.75	67.23	68.26	66.28	68.31	70.58
	± 0.38	± 0.22	± 0.39	± 0.07	± 0.17	± 0.10	± 0.45	± 0.02	± 0.35	± 0.10
EMO	87.17	71.07	59.32	84.25	67.89	66.09	69.04	62.26	66.47	70.40
	± 0.59	± 0.22	± 0.02	± 0.41	± 0.92	± 0.18	± 0.38	± 1.77	± 0.40	± 0.35
TALAS	92.44	75.58	65.13	87.09	85.39	64.39	78.14	72.56	81.03	77.97
	± 0.10	± 0.15	± 0.21	± 0.09	± 0.14	± 0.12	± 0.06	± 0.13	± 0.05	± 0.03
GATE-KD	92.56	76.30	65.62	86.58	87.20	63.43	79.23	72.69	80.60	78.24
	± 0.04	± 0.31	± 0.29	± 0.05	± 0.06	± 0.02	± 0.03	± 0.05	± 0.13	± 0.05
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)										
Teacher	92.51	71.28	65.80	83.38	90.09	66.84	80.65	77.11	84.59	79.14
Student (pre-KD)	67.03	69.45	42.90	78.39	64.55	59.58	47.60	21.96	22.08	52.62
RKD	89.74	72.34	57.99	84.13	75.20	67.49	74.48	68.12	72.55	73.56
	± 0.04	± 0.22	± 0.52	± 0.12	± 0.14	± 0.06	± 0.03	± 0.15	± 0.14	± 0.09
Stella	85.74	70.25	53.48	84.43	71.23	66.51	70.96	63.02	67.42	70.34
	± 0.22	± 0.07	± 0.28	± 0.04	± 0.05	± 0.14	± 0.28	± 0.04	± 0.24	± 0.02
CDM	85.18	69.14	52.30	84.02	68.24	66.57	69.15	63.05	66.55	69.36
	± 0.08	± 0.07	± 0.26	± 0.04	± 0.23	± 0.29	± 0.13	± 0.06	± 0.10	± 0.08
DSKD	85.04	69.21	52.85	84.34	67.33	65.98	69.01	63.46	66.42	69.29
	± 0.14	± 0.13	± 0.36	± 0.10	± 0.24	± 0.12	± 0.13	± 0.21	± 0.09	± 0.09
EMO	85.08	68.11	53.96	84.38	65.48	65.23	68.61	60.00	65.38	68.47
	± 0.17	± 0.10	± 0.60	± 0.16	± 0.16	± 0.17	± 0.07	± 0.82	± 0.38	± 0.15
TALAS	89.79	73.87	61.75	85.99	82.07	65.79	75.67	73.91	78.31	76.35
	± 0.03	± 0.17	± 0.61	± 0.03	± 0.09	± 0.04	± 0.05	± 0.03	± 0.04	± 0.08
GATE-KD	91.68	74.43	66.88	85.00	83.46	67.44	77.94	72.17	78.36	77.48
	± 0.09	± 0.21	± 0.36	± 0.06	± 0.02	± 0.04	± 0.06	± 0.02	± 0.07	± 0.04

Table 10: Training hyperparameters of the main experiments.

Method	Epochs	Batch size	Learning rate	Optimizer	Warm-up
RKD	5	128	7×10^{-5}	AdamW	0.06
Stella	2 + 3	128	5×10^{-5}	AdamW	0.06
CDM	5	128	2×10^{-5}	AdamW	0.06
DSKD	5	128	2×10^{-5}	AdamW	0.06
EMO	5	128	1×10^{-5}	AdamW	0.10
TALAS	5	128	2×10^{-5}	AdamW + adaptive SAM	0.06
GATE-KD	5	128	7×10^{-5}	AdamW	0.06